

Supplementation with vitamin C and N-acetyl-cysteine increases oxidative stress in humans after an acute muscle injury induced by eccentric exercise.

There has been no investigation to determine if the widely used over-the-counter, water-soluble antioxidants vitamin C and N-acetyl-cysteine (NAC) could act as pro-oxidants in humans during inflammatory conditions. We induced an acute-phase inflammatory response by an eccentric arm muscle injury. The inflammation was characterized by edema, swelling, pain, and increases in plasma inflammatory indicators, myeloperoxidase and interleukin-6. Immediately following the injury, subjects consumed a placebo or vitamin C (12.5 mg/kg body weight) and NAC (10 mg/kg body weight) for 7 d. The resulting muscle injury caused increased levels of serum bleomycin-detectable iron and the amount of iron was higher in the vitamin C and NAC group. The concentrations of lactate dehydrogenase (LDH), creatine kinase (CK), and myoglobin were significantly elevated 2, 3, and 4 d postinjury and returned to baseline levels by day 7. In addition, LDH and CK activities were elevated to a greater extent in the vitamin C and NAC group. Levels of markers for oxidative stress (lipid hydroperoxides and 8-iso prostaglandin F₂α; 8-Iso-PGF₂α) and antioxidant enzyme activities were also elevated post-injury. The subjects receiving vitamin C and NAC had higher levels of lipid hydroperoxides and 8-Iso-PGF₂α 2 d after the exercise. This acute human inflammatory model strongly suggests that vitamin C and NAC supplementation immediately post-injury, transiently increases tissue damage and oxidative stress.